

Objective-Driven Sequential Bayesian Experimental Design for Safe Learning and Control of Power-Grid Dynamics

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Abstract—Learning uncertain dynamic parameters is important for reliable power-grid operation. Existing experimental-design studies commonly maximize information gain, although greater parameter information does not necessarily improve a downstream control decision. This paper formulates power-grid probing as an objective-driven sequential Bayesian experimental-design problem. Each experiment selects a disturbance location and duration, observes the resulting frequency response, updates uncertainty over machine-specific dynamic parameters, and chooses a safety-aware control action. Expected information gain (EIG) and mean objective cost of uncertainty (MOCU) are studied as distinct design objectives. Fixed, random, myopic, deep adaptive design (DAD), reinforcement-learning sequential Bayesian optimal experimental design (RL-sBOED), and Step-DAD policies are evaluated under a common held-out protocol. A nonlinear susceptible–infected–recovered (SIR) model is used to validate the BOED implementations, while IEEE 9-bus and IEEE 14-bus systems evaluate control-oriented design, adaptivity, non-myopic planning, safety, and computational cost. [TODO: Insert final quantitative results after all seeds and validity checks are complete.] The results establish [TODO: insert the final evidence-supported practical conclusion].

Index Terms—Bayesian experimental design, power-system dynamics, adaptive experiment design, mean objective cost of uncertainty, information gain, safe control.

I. INTRODUCTION

INCREASING penetration of inverter-based resources makes power-system dynamic parameters more uncertain and potentially time varying. Reliable estimates of inertia, damping, and related dynamic quantities are important for frequency-security assessment and control selection. [TODO: Add citations on low-inertia grids and dynamic-parameter uncertainty.]

Active probing provides an opportunity to learn these quantities by applying a controlled disturbance and measuring the resulting frequency response. Such an experiment raises several design questions: where should the disturbance be applied, how long should it last, and should later experiments depend on earlier observations? A fixed sequence cannot respond to the realized system, whereas an adaptive policy can use the accumulating observation history.

Most Bayesian experimental-design formulations value an experiment through expected information gain. EIG is appropriate for parameter identification, but information that

reduces total parameter uncertainty is not necessarily the information most relevant to a downstream control decision. Different parameter values can imply the same control action, while small uncertainty in a decision-sensitive direction can materially change the required safe control. This distinction motivates objective-driven experimental design.

This work formulates power-grid probing as a sequential Bayesian experimental-design problem over discrete disturbance buses and durations. In addition to EIG, we consider a safety-aware MOCU objective that values an experiment by the expected reduction in the opportunity cost of the final control decision. The framework therefore connects physical probing, Bayesian posterior inference, and operational control within one evaluation protocol.

The main contributions are as follows:

- We formulate power-grid dynamic probing as sequential BOED over disturbance locations and durations.
- We define a control-oriented design objective based on safety-aware MOCU, connecting observations to a downstream safe control decision.
- We adapt and compare random, fixed, myopic, DAD, RL-sBOED, and Step-DAD under common banks, posterior inference, held-out systems, and evaluation noise.
- We investigate objective choice, adaptivity, non-myopic behavior, safety, horizon, observation noise, computational cost, and scalability on an SIR benchmark and IEEE power-grid systems.

[TODO: Add a one-paragraph roadmap after the manuscript sections stabilize.]

II. RELATED WORK

A. Bayesian Experimental Design

Introduce Bayesian optimal experimental design, expected utility, posterior inference, and EIG. Explain single-stage versus sequential design. [TODO: Add foundational BOED and EIG citations.]

B. Adaptive and Non-Myopic BOED

Discuss fixed/open-loop, myopic, and non-myopic policies. Introduce DAD as a fully amortized policy, RL-sBOED as a stepwise reinforcement-learning approach, and Step-DAD as semi-amortized online refinement initialized from DAD. [TODO: Cite the original papers and state precisely which parts are adapted.]

[TODO: Funding, corresponding author, and institutional acknowledgments.]

C. Objective-Driven Design and MOCU

Explain why task-specific uncertainty can be more useful than global parameter uncertainty. Review MOCU and related goal-oriented design criteria. [TODO: Add MOCU and objective-driven BOED citations.]

D. Power-Grid Identification and Active Probing

Review dynamic-parameter estimation, active probing, perturbation placement, PMU measurements, Bayesian uncertainty, and safe frequency control. End by identifying the gap addressed here: adaptive and non-myopic, control-oriented BOED over discrete bus-duration actions.

III. PROBLEM FORMULATION

A. Power-System Dynamics and Uncertainty

For each retained dynamic machine i , consider

$$\dot{\delta}_i(t) = \omega_i(t), \quad (1)$$

$$M_i \dot{\omega}_i(t) = P_{m,i} - P_{e,i}(\delta(t)) - K_i \omega_i(t) + p_i(t; d). \quad (2)$$

Here, δ_i and ω_i are rotor-angle and frequency deviations, M_i is inertia, K_i is a machine-specific damping or response coefficient, and $p_i(t; d)$ is the designed probe. Define

$$\theta = (M_1, \dots, M_G, K_1, \dots, K_G). \quad (3)$$

For IEEE14, dynamic machines are retained only at buses 1, 2, 3, 6, and 8; therefore, the uncertain vector is 10-dimensional. Non-dynamic buses are not assigned artificial M_i or K_i . [TODO: State and justify all prior ranges.]

B. Experimental Action Space

At step t ,

$$d_t = (b_t, \tau_t) \in \mathcal{D}, \quad (4)$$

where b_t is the disturbance bus and τ_t is its duration. With six durations and B buses, $|\mathcal{D}| = 6B$. IEEE9 therefore has 54 actions and IEEE14 has 84. [TODO: List the frozen, independently audited duration sets.]

C. Observation and Posterior Update

$$\mathbf{y}_t = h(\theta, d_t) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}). \quad (5)$$

The main power-grid protocol retains $N_{\text{obs}} = 5$ evenly spaced frequency-response samples. Define

$$h_t = \{(d_1, \mathbf{y}_1), \dots, (d_t, \mathbf{y}_t)\}. \quad (6)$$

Then

$$p(\theta | h_t) \propto p(\mathbf{y}_t | \theta, d_t) p(\theta | h_{t-1}). \quad (7)$$

D. Sequential Design Policy

$$d_t = \pi_t(h_{t-1}), \quad t = 1, \dots, T. \quad (8)$$

A policy is adaptive if later actions vary with observations. It is non-myopic if it accounts for effects on multiple future decisions rather than optimizing only the next immediate gain.

TABLE I
ROLES OF COMPARED METHODS

Method	Adaptive	Non-myopic	Learned
Random	No	No	No
Fixed	No	Open-loop	No
Myopic	Yes	No	No
DAD	Yes	Yes	Yes
RL-sBOED	Yes	Yes	Yes
Step-DAD	Yes	Yes	DAD + refine

IV. DESIGN OBJECTIVES

A. Expected Information Gain

$$\text{EIG}(d | h) = \mathbb{E}_{\mathbf{y}|d,h} [D_{\text{KL}}(p(\theta | h, d, \mathbf{y}) \| p(\theta | h))]. \quad (9)$$

EIG values parameter learning and does not directly encode the final control decision. Higher EIG is better.

B. Safety-Aware Control Decision

Let $u \in \mathcal{U}$ denote a control magnitude and $C(u, \theta)$ its safety-aware cost. Define

$$u^*(\theta) = \arg \min_{u \in \mathcal{U}} C(u, \theta), \quad (10)$$

$$u^*(h_t) = \arg \min_{u \in \mathcal{U}} \mathbb{E}_{\theta|h_t} [C(u, \theta)]. \quad (11)$$

[TODO: State the frequency-security constraint, control grid, safety quantile, and penalties exactly as implemented.]

C. MOCU Objective

$$\text{OCU}(\theta, h_t) = C(u^*(h_t), \theta) - C(u^*(\theta), \theta), \quad (12)$$

$$\text{MOCU}(h_t) = \mathbb{E}_{\theta|h_t} [\text{OCU}(\theta, h_t)]. \quad (13)$$

Lower MOCU is better. MOCU equal to zero means posterior uncertainty creates no regret relative to the clairvoyant control. The central objective is

$$\pi^* = \arg \min_{\pi} \mathbb{E}_{\pi} [\text{MOCU}(h_T)]. \quad (14)$$

V. COMPARED DESIGN METHODS

A. Random Design

Random selects feasible actions without using the history and is averaged over repeated random sequences.

B. Fixed Open-Loop Design

Fixed chooses the complete sequence before observing data,

$$d_{1:T}^{\text{fixed}} = \arg \min_{d_{1:T}} \mathbb{E} [\text{MOCU}(h_T)]. \quad (15)$$

The same sequence is used for every system. Observations update the posterior and final control but cannot alter later experiments. ‘‘Fixed MOCU’’ is the same terminal metric achieved by the fixed policy, not a separate objective.

C. Myopic Design

Myopic recomputes the best immediate action after each observation but does not explicitly value later design opportunities.

D. Deep Adaptive Design

DAD trains a history-conditioned policy offline, making online selection fast. [TODO: Describe policy inputs, architecture, estimator, training objective, and fidelity to the original paper.]

E. RL-sBOED

RL-sBOED trains a history-conditioned sequential policy with stepwise rewards. [TODO: Describe the telescoping reward, optimizer, and discrete-action changes.]

F. Step-DAD

Step-DAD initializes from DAD and performs limited history-specific online refinement. It trades online time for a possible reduction in DAD's amortization gap. Report DAD offline training, refinement, and total cost separately. [TODO: Include only after matched experiments pass validation.]

VI. EXPERIMENTAL PROTOCOL

A. SIR ODE Benchmark

The nonlinear SIR benchmark validates the EIG implementation in a simpler setting. Its action is an observation time and its scalar response is infected population $I(t)$. SIR is a validation benchmark, not the application novelty.

B. IEEE Test Systems

C. Method-Independent Duration Audit

Before training, duration subsets are screened for positive adaptive and non-myopic value, at least two meaningful duration branches, and limited branch dominance. The audit must use exactly the same MOCU, safety quantile, control grid, observation count, and noise as training and evaluation. Durations are not selected to make a particular method win.

D. Training and Held-Out Evaluation

Use training seeds 101, 202, and 303. All methods share the physical bank, prior support, held-out systems, observation-noise seed, and terminal metric. Learned policies use deterministic argmax actions during evaluation. A method is valid only if held-out safety meets the frozen threshold. [TODO: State held-out sample count, replicates, hardware, software, and stopping rules.]

VII. RESULTS

A. Implementation Validation on SIR

[TODO: Insert SIR EIG-versus-horizon results with mean, standard deviation, paired comparisons, adaptive sequences, and runtime.]

B. Effect of Objective Choice on IEEE9

[TODO: Compare EIG-driven and MOCU-driven experiments. Explain whether greater EIG implies lower downstream MOCU.]

C. Adaptive and Non-Myopic Performance

Table III is the primary table. Report mean and sample standard deviation over three independent training seeds. Lower values are better. Do not bold values until safety and statistical checks are complete.

D. Adaptivity Diagnostics

Report unique sequences, sequence entropy, meaningful posterior-dependent branches, and dominant branch share. A nominally adaptive policy producing one sequence is operationally open-loop and is not evidence of adaptation.

E. Safety and Control Performance

Report mean MOCU, mean selected control, under-control frequency, and safety rate. Methods below the safety threshold receive no rank.

F. Computational Cost

Separate bank generation, offline training, and online action selection. For Step-DAD, also report posterior fitting and policy refinement.

G. Scalability

Compare IEEE9, IEEE14, and, if completed, IEEE30 using latent dimension, action count, bank size, training and online time, safety, and terminal MOCU.

H. Robustness and Ablations

Use a prespecified study of observation noise or observation count. Avoid changing multiple factors simultaneously. [TODO: Specify the frozen matrix.]

VIII. STATISTICAL ANALYSIS

For $S = 3$ training seeds, report

$$\bar{x} = \frac{1}{S} \sum_{s=1}^S x_s, \quad (16)$$

$$s_x = \sqrt{\frac{1}{S-1} \sum_{s=1}^S (x_s - \bar{x})^2}. \quad (17)$$

On common held-out systems, compute paired differences

$$\Delta_i = \text{MOCU}_{A,i} - \text{MOCU}_{B,i}. \quad (18)$$

For MOCU, $\Delta_i < 0$ favors A . Report the paired mean, 95% confidence interval, and win fraction. Separate training-seed variability from held-out system variability. Use “statistically significant” only after a defined test.

TABLE II
SYSTEM AND EXPERIMENTAL-DESIGN CONFIGURATION

System	Physical buses	Dynamic buses	Latent dim.	Actions	N_{obs}	σ
SIR ODE	—	—	[TODO: 2]	[TODO: 100 times]	1	[TODO: value]
IEEE9	9	3	6	54	5	0.005
IEEE14	14	5	10	84	5	0.005
IEEE30	30	[TODO:]	[TODO:]	[TODO:]	5	0.005

TABLE III
TERMINAL SAFETY-AWARE MOCU (MEAN $\pm$ STANDARD DEVIATION)

Method	$T = 3$	$T = 4$	$T = 5$	$T = 6$	$T = 7$
Random	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Fixed	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Myopic	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
DAD	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RL-sBOED	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Step-DAD	[TODO: optional]	[TODO: optional]	[TODO: optional]	[TODO: optional]	[TODO: optional]

TABLE IV
ADAPTIVITY DIAGNOSTICS AT A REPRESENTATIVE HORIZON

Method	Unique seq.	Seq. entropy	Dominant share
Fixed	1	0	1.00
Myopic	[TODO:]	[TODO:]	[TODO:]
DAD	[TODO:]	[TODO:]	[TODO:]
RL-sBOED	[TODO:]	[TODO:]	[TODO:]
Step-DAD	[TODO:]	[TODO:]	[TODO:]

TABLE V
COMPUTATIONAL-COST TEMPLATE

Method	Offline train	Online/action	Deployment total
Fixed	[TODO:]	[TODO:]	[TODO:]
Myopic	—	[TODO:]	[TODO:]
DAD	[TODO:]	[TODO:]	[TODO:]
RL-sBOED	[TODO:]	[TODO:]	[TODO:]
Step-DAD	[TODO: DAD]	[TODO: refine]	[TODO:]

IX. DISCUSSION

Organize the discussion around these questions:

- 1) When does observation-dependent adaptation improve control?
- 2) When is a strong fixed design competitive?
- 3) When does non-myopic planning outperform myopic selection?
- 4) Does EIG improvement translate to lower MOCU?
- 5) Does increasing T cause information or decision saturation?
- 6) Is extra online computation justified by improvement?
- 7) How do action count and latent dimension affect scalability?

A negative result is informative. For example, a competitive fixed design can indicate limited posterior-dependent decision branching.

X. LIMITATIONS AND FUTURE WORK

The study uses simulated IEEE systems, prescribed priors, Gaussian noise, fixed probe amplitude, discrete actions, and a

simplified downstream control. Three training seeds provide a limited estimate of optimizer variability. Future work should consider larger systems, correlated and time-varying priors, multiple amplitudes, PMU channel selection, higher-fidelity constraints, and real or hardware-in-the-loop validation.

XI. CONCLUSION

This paper formulated power-grid probing as objective-driven sequential BOED over disturbance buses and durations. The framework connected Bayesian learning to safety-aware control through terminal MOCU and compared fixed, myopic, adaptive, and non-myopic policies under a common protocol. [TODO: Insert only final three-seed findings, including cases where fixed or myopic remained competitive.] These findings clarify [TODO: state when adaptive objective-driven probing is practically worthwhile].

APPENDIX A

REPRODUCIBILITY CHECKLIST

- Frozen YAML configuration for every reported experiment.
- Physical-bank and control-bank hashes.
- Train, validation, test, and evaluation-noise seeds.
- Duration-audit settings and gates.
- Policy architecture and optimizer settings.
- Hardware and software versions.
- Per-seed summaries and paired held-out rollouts.
- Exact commands for every table and figure.

APPENDIX B

SUGGESTED FIGURE PLAN

- 1) Prior $\rightarrow$ probe $\rightarrow$ observation $\rightarrow$ posterior $\rightarrow$ control workflow.
- 2) IEEE bus-duration action-space illustration.
- 3) EIG versus horizon on SIR and IEEE9.
- 4) MOCU versus horizon on IEEE9 and IEEE14.
- 5) Safety rate versus horizon.
- 6) Observation-conditioned branching examples.
- 7) Offline and online runtime comparison.
- 8) Performance and cost versus system size.

ACKNOWLEDGMENT

[TODO: Add funding, computing resources, and individual contributions.]

REFERENCES

- [1] [TODO: Add verified references for BOED, DAD, RL-sBOED, Step-DAD, MOCU, power-grid probing, and dynamic-parameter estimation.]